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A dispersive optical model for $n + {}^{120}$Sn from −15 to +80 MeV and properties of neutron single-particle and single-hole states

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Abstract

Data for $\sigma(\theta)$ and $A_y(\theta)$ previously obtained at the Triangle Universities Nuclear Laboratory for ^{120}Sn(n, n) are combined with other measurements of $\sigma(\theta)$ and $A_y(\theta)$ to create an elastic-scattering database from 9.9 to 24 MeV. In addition, relatively recent high-accuracy measurements of the neutron total cross section σ_T for Sn from 5 to 80 MeV are combined with earlier σ_T data to form a detailed σ_T database from 0.24 to 80 MeV. All of these data are analysed in the framework of a dispersive optical model (DOM). The DOM is extended to negative energies to investigate properties of single-particle and single-hole bound states. The DOM also is used in calculations of compound-nucleus contributions to $\sigma(\theta)$, so that DOM predictions can be compared to $\sigma(\theta)$ measurements. Excellent agreement is obtained for the entire set of scattering data from 0.4 to 24 MeV, and for σ_T values from 0.05 to 80 MeV. Calculations of bound-state quantities are compared to values derived from experiment for energies down to −15 MeV. Reasonable agreement for the binding energies is achieved, while the predicted spectroscopic factors disagree somewhat with the values found in stripping and pickup experiments. Finally, the DOM is modified to investigate two features (volume absorption that is asymmetric about the Fermi energy and zero absorption in the vicinity of the Fermi energy) that have been ignored in many DOM models. These modifications have little effect on the agreement of the calculations with the scattering data or with the bound-state quantities.

1. Introduction

The dispersive optical model (DOM) has been used to describe nucleon–nucleus scattering for about ten nuclei in the last 15 years. The success of the description has been verified through comparisons with measured data for total cross-section σ_T and elastic-scattering differential cross-section $\sigma(\theta)$ and analysing power $A_y(\theta)$ over wide energy ranges. The validity of the DOM approach has also been confirmed in comparisons with experimentally determined binding energies and spectroscopic factors of single-particle and single-hole states. The success of this approach to explain observed bound-state features was first illustrated by Johnson *et al* [1], who calculated the binding energies, occupation probabilities and spectroscopic factors of the neutron single-particle and single-hole states for the $n + {}^{208}$Pb system. Their results fell between those obtained in nuclear matter calculations and those determined from pickup and stripping experiments. Other successful DOM studies followed, for systems such as $n + {}^{90}$Zr [2] and $n + {}^{40}$Ca [3].

At TUNL we have developed a computer code [4, 5] that allows us to characterize a DOM using a search routine for minimizing χ^2 for a large set of $\sigma(\theta)$, $A_y(\theta)$ and σ_T data for neutron scattering. Our DOM includes modifications suggested in the work of Mahaux and Sartor [6] that set the absorption potential to zero in a few-MeV region centred on the Fermi energy (E_F) and to allow the absorption to be asymmetric about E_F. In addition, we have included corrections for relativistic effects in order to compare relatively recent high-accuracy σ_T data [7] over the energy range from 5 MeV to 80 MeV.

In the present work, we used our search code to create a DOM for the $n + {}^{120}$Sn system from 0 to 80 MeV. The model gives a good description of the available scattering data for $\sigma(\theta)$ and $A_y(\theta)$ from 0.4 to 24 MeV and for σ_T data over most of the energy range from 0.05 to 80 MeV. For some applied purposes, it might be that our DOM potential gives adequately reliable predictions of differential cross sections and analysing powers over the entire energy range up to 80 MeV. Subsequently, we extrapolated our DOM to the negative energy regime and used it to predict bound-state properties for neutron single-particle and single-hole states for energies down to -15 MeV. Most of our predictions of binding energies agree within about 10% of the reported values, while our predictions of spectroscopic factors agree within about 35% of the reported values.

2. Database

Two sets of experimental data were collected from the literature. A subset, the 'search database', was the database employed in the DOM search procedure. A larger set, the 'test database' was a combination of the search database with a large amount of additional differential and total cross-section data. This larger set was used in the later stages of the development to compare more fully the measured data to DOM predictions.

For the search database we chose data at high enough energies to avoid interference from compound nucleus contributions, starting with the TUNL $\sigma(\theta)$ data of Guss *et al* [8] at 9.94, 13.92 and 16.91 MeV for ^{120}Sn. We also used the $\sigma(\theta)$ data of Rapaport *et al* [9] at 11.0 MeV for ^{120}Sn and at 24.0 MeV for ^{118}Sn. For $A_y(\theta)$, we included the TUNL data of Guss *et al* [8] at 9.94 and 13.92 MeV for ^{120}Sn and Pedroni *et al* [10] at 16.91 MeV for ^{120}Sn. We also used the $A_y(\theta)$ data of Wong *et al* [11] at 23.7 MeV for ${}^{\mathrm{nat}}$Sn. The σ_T data for our search are from: Finlay *et al* [7] at 18 energies between 5.5 and 80 MeV for ${}^{\mathrm{nat}}$Sn; Poenitz and Whalen [12] at five energies between 1.60 and 4.91 MeV for ${}^{\mathrm{nat}}$Sn; and Timokhov *et al* [13] at four energies between 0.24 and 1.25 MeV for ^{120}Sn.

For the 'test database' we included all the above data plus $\sigma(\theta)$ data from: Musaelyan *et al* [14] at 0.4, 0.6, 0.8, 0.93, 1.06, 1.45 and 1.96 MeV for ^{120}Sn; Tanaka *et al* [15] at 1.52, 2.05, 2.57, 3.08, 3.60 MeV for ^{120}Sn; and Smith *et al* [16] at 5.00, 6.50, 8.08 and 9.99 MeV for ${}^{\mathrm{nat}}$Sn. In addition, the predictions based on our DOM are compared to σ_T data for 272 values from Finlay *et al* [7] for ${}^{\mathrm{nat}}$Sn from 5.29 to 80 MeV, for 68 values from Poenitz and

Whalen [12] for ^{nat}Sn from 0.047 to 5.273 MeV, and for 64 values from Timokhov *et al* [13] for ^{120}Sn from 0.022 to 1.35 MeV.

In order to have an adequate database for the search, we used a mixture of data for the different isotopic compositions, including that for the naturally occurring abundances ^{nat}Sn (for which we used a mass of 118.70 u). In addition, the DOM developed here did not use an explicit isospin term. To deal with these omissions, we performed two initial investigations. In the first step we developed (with the aid of the search procedure) a DOM for the case where each measured data set carried the correct isotopic mass, thereby forcing the potential radii to be roughly proportional to $A^{1/3}$. Using the obtained DOM, we calculated the observables. In a second preliminary step we fixed all the isotopic masses at that for ^{120}Sn and inserted the real isospin term obtained from the global potential of [17] into the real potential of the DOM and calculated the observables.

As expected, we found that the radius and isospin effects are small and partially compensate one another. For example, if ^{118}Sn data are being considered, then using the mass for ^{120}Sn (119.90 u) slightly strengthens the DOM potential through the $A^{1/3}$ dependence. However, the lack of an isospin term leaves the potential shallower than it should be (since the isospin term would make the potential for ^{118}Sn slightly deeper than that for isotopes containing a greater number of neutrons). The resulting potentials and calculated observables showed insignificant differences. Only the values for σ_T, which have been measured to an absolute accuracy of $\pm 1\%$, changed by an amount equal to an appreciable fraction of the error bar of the measurements. Since we were primarily interested in developing a DOM to study $n + ^{120}$Sn, we neglected the isospin term and mass differences for all the data and took the mass to be that for ^{120}Sn.

3. Form of the dispersive optical model

The nucleon–nucleus optical potential $U(E)$ contains real and imaginary terms: $U(E) = V(E) + \mathrm{i}W(E)$. The dispersion relation, which connects the imaginary term to the real term, generates a correction term $\Delta V(E)$ comprised of two pieces, one having a volume shape and one having a surface shape. Thus, we obtain $U(E) = V_{\mathrm{HF}}(E) + \Delta V_v(E) + \Delta V_s(E) + \mathrm{i}W(E)$. Here, $V_{\mathrm{HF}}(E)$ is the Hartree–Fock term which represents the nuclear mean field. In a more complete representation we obtain the optical potential as:

$$\begin{aligned} U(r, E) = {} & -[V_{\mathrm{HF}}(E) + \Delta V_v + \mathrm{i}W_v(E)] f_v(r, R_v, a_v) - [\Delta V_s(E) + \mathrm{i}W_s(E)] g_s(r, R_s, a_s) \\ & - \left[V_{\mathrm{so}}(E) g_s\left(r, R_{V_{\mathrm{so}}}, a_{V_{\mathrm{so}}}\right) + \mathrm{i}W_{\mathrm{so}}(E) g_s\left(r, R_{W_{\mathrm{so}}}, a_{W_{\mathrm{so}}}\right)\right] \left(\frac{\hbar}{m_\pi c}\right)^2 \frac{1}{r}(\boldsymbol{\ell} \cdot \boldsymbol{\sigma}) \end{aligned} \tag{1}$$

where the subscripts v, s and so indicate the volume, surface and spin–orbit components, respectively. The $f_v(r, R_i, a_i)$ and $g_s(r, R_i, a_i)$ are the volume and surface form factors, respectively, and are conventional Woods–Saxon forms: $f_v(r, R_i, a_i) = [1 + \exp(r - R_i/a_i)]^{-1}$ and $g_s(r, R_i, a_i) = -4a_i\, \mathrm{d}/\mathrm{d}r[f_v(r, R_i, a_i)]$, where $R_i = r_i A^{1/3}$. From equation (1) one sees that we force the radii and the diffuseness of the real and imaginary volume terms to have the same value; this constraint simplifies the combination of the dispersive contribution ΔV_v with the Hartree–Fock component V_{HF}. In our potential, r_i and a_i are independent of energy.

The dispersion contribution to the real potential from the absorption term is given by [1]

$$\Delta V_j(E) = \frac{P}{\pi} \int_{-\infty}^{+\infty} \frac{W_j(E')}{E' - E}\, \mathrm{d}E' \tag{2}$$

where P stands for the principal value, and $j = v$ and $j = s$ represent the volume and surface contributions, respectively.

For the central real potential V_{HF} we chose the energy dependence [18]:

$$V_{\mathrm{HF}}(E) = A_{\mathrm{HF}} \exp\left(-\frac{B_{\mathrm{HF}}}{A_{\mathrm{HF}}}(E - E_F)\right) \tag{3}$$

where the parameters A_{HF} and B_{HF} are constants to be determined in the analysis. To determine the value of the Fermi energy E_F we followed the convention of Mahaux *et al* [19] in which E_F lies midway between the first particle and first hole state for neutrons in ^{120}Sn. That is, where E_F is the average of the separation energy of a neutron from ^{121}Sn and the separation energy of a neutron from ^{120}Sn. For the nucleus ^{120}Sn this prescription gives $E_F = -(1/2)(6.171 + 9.107)$ MeV $= -7.639$ MeV.

The volume absorptive term W_v uses the energy dependence reported in [20]:

$$W_v(E) = \frac{A_v(E - E_F)^n}{(E - E_F)^n + (B_v)^n} \tag{4}$$

where n is to be determined through fitting the data. We investigated $n = 2$, 4 and 6. For the surface term W_s we utilized the energy dependence that was successfully employed initially by Delaroche *et al* [2] for ^{90}Zr:

$$W_s(E) = \frac{A_s(E - E_F)^m}{(E - E_F)^m + (B_s)^m} \exp(-C_s|E - E_F|). \tag{5}$$

Here we tested the values $m = 2$ and 4.

One difference from the early DOM work is due to the recognition of Mahaux and Sartor [21] that the absorption in the immediate vicinity of E_F should be zero. In a study of $n + {}^{208}$Pb, they calculated the average energy $|E_p|$ of the bound particle states (for $E > E_F$) and forced W_v to go to zero inside the region from $E = E_F - |E_p|$ to $E = E_F + |E_p|$ by shifting E by E_p in equations (4) and (5). We followed this approach; the results of testing several values for E_p are mentioned below.

In the early stages of our analysis of ^{120}Sn both $W_v(E)$ and $W_s(E)$ were assumed to go to zero only at exactly $E = E_F$ and to be symmetric about $E = E_F$; this approximation is used in most DOM work. In one of their later papers Mahaux and Sartor [6] introduced an asymmetric form about E_F for the volume absorption $W_v(E)$ into the dispersion-relation integrand to account for non-locality effects. The corrections are applied for E greater than 60 MeV above E_F and for E less than 60 MeV below E_F. That is, $W_v(E)$ is taken to be that in equation (4) only for $(E_F - 60) < E < (E_F + 60)$ MeV. The following approximations were given for $E < (E_F - 60)$ MeV:

$$\mathcal{W}_v(E) = W_v(E) - W_v(E_F - 60)\frac{(E - E_F + 60)^2}{(E - E_F + 60)^2 + 60^2} \tag{6}$$

and for $E > (E_F + 60)$ MeV:

$$\mathcal{W}_v(E) = W_v(E) + \alpha\left[E^{1/2} + \frac{(E_F + 60)^{3/2}}{2E} - \frac{3}{2}(E_F + 60)^{1/2}\right]. \tag{7}$$

Here $\mathcal{W}_v$ denotes the non-local imaginary volume potential. The calculations in [6] led to the value of $\alpha = 1.65$. An extension of this work [22] raises concerns about the approximations underlying this treatment of non-locality and the particular determination of α. In light of these concerns, we attempted to optimize our DOM by using a variety of values for α, including setting α equal to zero, which means applying the non-locality corrections only to the region of $E < (E_F - 60)$ MeV. The results are discussed below.

Mahaux and Sartor [6] argue that the effect of non-locality in the W_s term is not significant because the magnitude of the surface absorption ought to drop to zero in the vicinity of 60–100 MeV. We accepted this view and used the symmetric form, equation (5), for W_s throughout our study.

4. The search procedure

As described in [4, 5], we modified the global search code GENOA (obtained from F Perey of ORNL) so that it incorporated the DOM formalism and analytical solutions of equation (2), the dispersion relation. GENOA minimizes the total χ^2 where χ is the difference between the data point and the value calculated from the model. Prior to calculating the total χ^2, the χ^2 value for a particular data set was multiplied by a weighting factor K_i. This factor allowed us to give either more or less weight to data sets in a particular energy range or to particular types of data. For instance, the TUNL $A_y(\theta)$ data were given double weighting because there were only three complete angular distributions of $A_y(\theta)$ available and because we wanted to constrain the spin–orbit parameters through polarization data. To speed the search, σ_T values were used only at 27 energies. In order for these σ_T values to carry moderate influence during the fitting procedure, they were given about ten times the weighting factor of a typical $\sigma(\theta)$ data set. The impact of this heavier weight was particularly significant above 24 MeV where there were no differential data in our database.

The initial parameters for the search were taken from global potential models and from our early (unpublished) DOM work for Sn isotopes. First, the range of parameter space for most of the parameters was investigated and sensitivity studies were done for certain parameters by a grid search. In the final stages, certain parameters were held fixed while others were optimized by allowing GENOA to minimize the χ^2. Periodically, the weighting factors for various data sets were adjusted to allow better overall qualitative fits across the database.

Before proceeding to our final results, we will describe the constraints placed on the DOM parameters before and during the search. The spin–orbit interaction in equation (1) contains both a real and an imaginary term, as well as energy-dependent strengths. The presence of the imaginary term in the neutron–nucleus spin–orbit interaction was not widely accepted at the time of our initial studies. However, after many optical model-investigations on nuclei other than Sn that were conducted in previous TUNL analyses of TUNL $A_y(\theta)$ measurements, it was concluded that the quality of the fits to the $A_y(\theta)$ data at forward angles is substantially improved by the inclusion of a W_{so}. In followup studies with a conventional spherical optical model, we found that to be true also for Sn. For these reasons, we incorporated an imaginary spin–orbit term in the present analysis. In regard to the energy dependences of $V_{\mathrm{so}}(E)$ and $W_{\mathrm{so}}(E)$, the present energy range (9.94–16.9 MeV) of the highest quality $A_y(\theta)$ data is too limited to establish meaningful slopes. In the present DOM work we initiated the search with the linear energy dependences and the slopes for $V_{\mathrm{so}}(E)$ and $W_{\mathrm{so}}(E)$ found in the global conventional optical model reported by Walter and Guss [17]. As compared to [17], the final values reported below have the same linear slopes and have strengths within a few hundred keV.

Values of $n = 2$, 4 and 6 were investigated in equation (4), and $m = 2$ and 4 in equation (5). The $n = 6$ calculations were in close agreement with those for $n = 4$. The best results for the overall database were obtained with $n = 4$ and $m = 2$, so we adopted these values.

As was mentioned previously, we adopted the simplifying assumption of [1] that the radius r_{Vv} of the real central potential is the same as the radius r_{Wv} of the volume imaginary potential. We made the same assumption for the diffuseness a_{Wv}. Although this does make it convenient to add the volume dispersive correction to that of the real central term, it does force a compromise on the quality of the fits, since it is common in conventional optical models for r_{Wv} to be about 10% larger than r_{Vv}. Regardless, as we show later, the comparisons between the Sn data and calculations are very good. (In a DOM study for $n + {}^{208}$Pb, we decoupled these radii and proved that for ^{208}Pb the improvement in the quality of the fits is insignificant [23].)

On the other hand, the optimum fits preferred a larger value for r_{Vv} than is normally found in optical model analyses. To study this in detail, the r_{Vv} was stepped from 1.230 to 1.245 fm. The minimum χ^2 was observed at the largest value. We decided to keep the value closer to the region found in other optical models, and to not allow the value to exceed 1.240 fm, so we fixed it at this value. In addition, the minimization of χ^2 favoured an unacceptably large value for the surface absorption radius, that is, $r_{Ws} = 1.29$ fm. We were able to force this parameter to smaller values, eventually reaching 1.240 fm without losing good agreement with the data.

The form of the absorption potential in equation (5) is quite elementary. In fact, it might be too simple to properly describe the interaction over the entire energy range from -15 to 80 MeV. The coefficient C_s is the determining factor for the fall-off of the surface absorption at high energies. In most optical models the surface absorption drops to zero around 50–80 MeV. In microscopic models, such as the successful one of Jeukenne *et al* [24], the absorption is predominantly volume beyond 80 MeV. Our fits to the Sn database favoured a value for $C_s = 0.018$, a value that is too small to drop the surface absorption fast enough with increasing energy to be consistent with earlier optical models. Since our model at high energies is primarily constrained by fitting the total cross section, and since this observable is insensitive to a large range of combinations of volume and surface absorption, this low value of C_s might be compensating for some fine details in the low-energy data sets. Because of this uncertainty we decided to force the surface absorption term to lower values at high energies; we therefore fixed C_s at the value of 0.045.

We investigated the insertion of the offset energy E_p that determines the range ΔE of energy around E_F for which the absorption is zero. Tests were made with offset values from 1 to 4 MeV, i.e., $\Delta E = 2$ to 8 MeV. The value of the minimum in χ^2 was very insensitive across this range. We fixed E_p at 2 MeV, which makes the ΔE gap to be about 30% greater than the gap between the first hole and the first particle bound states.

At periodic intervals in the search procedure, the sensitivity of bound-state properties to parameter changes was tested. Also, calculated bound-state values were compared to measured values. The DOM parameters were shifted slightly in the direction that gave better agreement. Then the search on the cross-section and analysing power data was continued to seek convergence.

5. Compound-nucleus calculations

For the DOM search, only differential data for $E \geqslant 9.94$ MeV were included. The compound-nucleus (CN) contribution to the elastic scattering is negligible in this energy range, so the conclusions drawn from the DOM search are unaffected by CN effects. However, to justify using the DOM for predicting bound-state properties and for applied purposes, we checked that the model gives a reasonable description for elastic scattering below 9.9 MeV. To accomplish this, it was necessary to first combine the contribution for compound-elastic scattering $\sigma^{CN}(\theta)$ with the direct-reaction scattering $\sigma(\theta)$ calculated from the DOM.

The $\sigma^{CN}(\theta)$ was calculated using the code OPDISP that was developed [26] at Ohio University by modifying the Hauser–Feshbach code OPSTAT. The revised code allows the

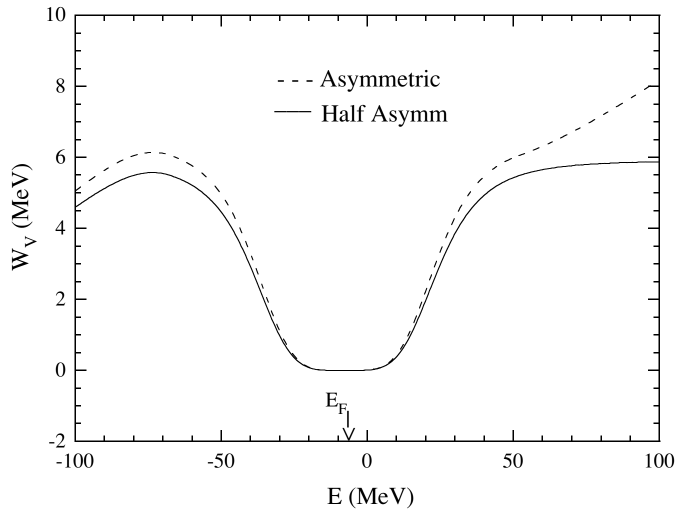

Figure 1. Values of the volume absorption obtained from searching on scattering and total cross-section measurements, for case ii, the asymmetric W_v (dotted line), and case iii, the 'half asymm' W_v (solid). Values for case i, the symmetric W_v, can be obtained by reflecting the solid curve from $E = E_F$ to +100 MeV about the Fermi energy E_F.

calculation of penetrabilities using a DOM, but only with a W_v and a W_s that are symmetric about E_F. We used this code for all the DOM CN calculations reported, even for the models with the asymmetric W_v, in which cases we neglected the asymmetric factor. Because the non-locality corrections have their greatest influence on the dispersive contributions above 20 MeV and because CN contributions were significant only for neutron energies below 4 MeV, we believe that this is a valid choice.

For the CN calculations we also assumed that the only important channels were elastic and inelastic neutron scattering. We followed the method of Moldauer [27] to deal with the width-fluctuation correction factor and set $W_{\alpha\beta} = 2$ for the elastic channel. In addition to the calculations for the comparisons to the elastic scattering data shown in the next section, we tested our model by calculating the inelastic differential cross section for the first excited state of ^{120}Sn at $E_n = 1.63$ MeV and $\theta = 90°$. Our value of 37 mb/sr agrees with the measured 38 mb/sr reported by Harper *et al* [28].

6. Results and comparison to the scattering and total cross-section data

Three sets of calculations were carried out in the later stages of the search. These were with a DOM for case i, a symmetric W_v; case ii, the Mahaux and Sartor asymmetric W_v, as in equations (6) and (7); and case iii, an intermediate model (denoted as 'half asymm') between these two in which the non-locality correction to W_v was applied only in the energy range below E_F. Case iii was motivated by the fact that no comfortable solution to the differential and total cross-section data could be found in case ii. We found that if equation (6) was used but not equation (7), then the good agreement obtained with a symmetric W_v did not deteriorate. The energy dependences of the strengths of W_v for cases i, ii and iii are shown in figure 1. (For illustration, we graph these from -100 MeV to +100 MeV, even though we only investigated predictions in the range from $-15 < E < 80$ MeV.)

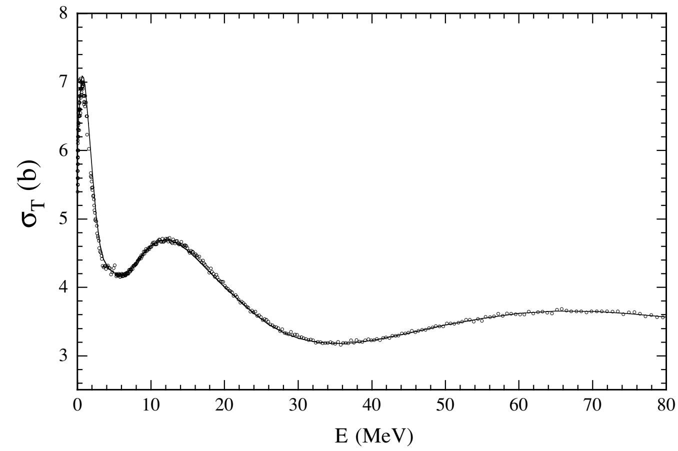

Figure 2. Comparison of total cross-section values (circles) for $n + \mathrm{Sn}$ from several experiments to values (solid curve) predicted with the DOM. (See text for sources of data.) Most of the data shown here are from the 'test database.' Note the offset of the ordinate from 0.0.

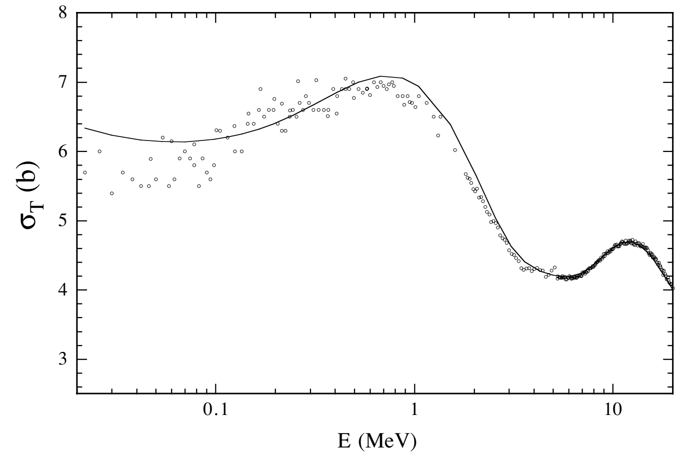

Figure 3. Comparison of total cross-section values (circles) for $n + \mathrm{Sn}$ from 0.020 to 20 MeV to values (solid curve) predicted with the DOM. (See caption of figure 2.)

Table 1. DOM parameters for $n + {}^{120}\mathrm{Sn}$ scattering (case i).

A_{HF} (MeV)	47.876	a_s (fm)	0.544
B_{HF}	0.3609	V_{so} (MeV)	5.956
r_v (fm)	1.240	$\alpha_{V_{\mathrm{so}}}$	−0.015
a_v (fm)	0.666	$r_{V_{\mathrm{so}}}$ (fm)	1.179
A_v (MeV)	5.9177	$a_{V_{\mathrm{so}}}$ (fm)	0.607
B_v (MeV)	30.529	W_{so} (MeV)	0.915
A_s (MeV)	25.484	$\alpha_{W_{\mathrm{so}}}$	−0.018
B_s (MeV)	13.905	$r_{W_{\mathrm{so}}}$ (fm)	1.267
C_s (MeV^{-1})	0.045	$a_{W_{\mathrm{so}}}$ (fm)	0.594

For the DOM of case i, which used the symmetric form of W_v, we list the DOM parameters in table 1 and display the model predictions in figures 2–6. The calculations for σ_T are shown

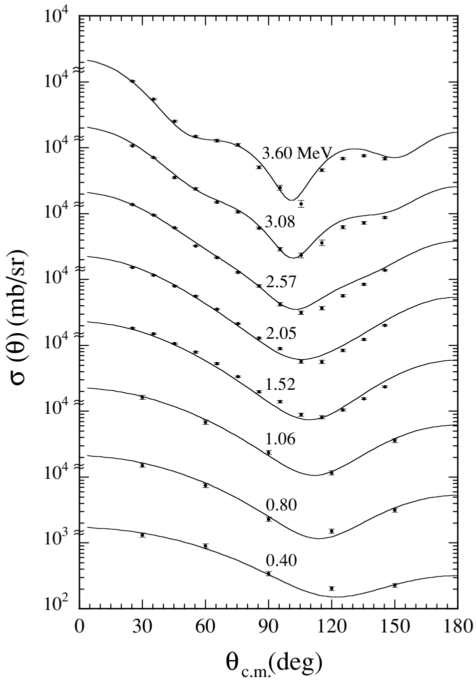

Figure 4. Plot of measured differential cross-section values for Sn(n, n) (solid circles). (See text for sources.) The solid curves are the values obtained by adding the compound nucleus calculation to the DOM calculation. The data are from the 'test database'.

in figures 2 and 3. The absolute error bars on σ_T measurements in the 5–80 MeV region are about $\pm 1\%$. The DOM calculations give excellent agreement with the σ_T data. (Note the zero offset along the ordinate.) Recall that only 27 of the σ_T values from 0.24 to 80 MeV were used in the search along with differential data from 9.9 to 24 MeV. The $\sigma(\theta)$ data and calculations for the symmetric case are shown in figures 4 and 5. For 3.60 MeV and below, the contributions for $\sigma^{CN}(\theta)$ have been added to the $\sigma(\theta)$ calculated with the DOM. Overall, the agreement for $\sigma(\theta)$ is very good; it is possible that some of the discrepancies might be attributed to systematic errors in the measurement, for instance, errors associated with the scattering-angle calibration or with the absolute normalization errors. (Systematic errors play havoc in a chi-squared search routine; this is one reason why we did not weight all data sets equally.) All the available $A_y(\theta)$ data for $E_n \geqslant 9.94$ MeV are compared to the calculations in figure 6. Considering the size of the error bars, these data are also quite well represented by the calculations.

For cases ii and iii, we do not display our DOM parameters or predictions, but only give a brief discussion of the results. For case ii, which used the asymmetric W_v of equations (6) and (7), the weighted total χ^2 was worse by 5% compared to that of case i. The primary discrepancy between the calculations of case ii and data was in the description of the total cross section, where the calculated value is about 1.4% below the data at 60 MeV and about 1.0% above the data at 80 MeV. It is clear that the model of case ii is deficient, at least when all the geometry parameters are constrained to be independent of energy.

The results for case iii, using the 'half asymm' W_v potential of equation (6) only, gave predictions that were indistinguishable from case i. The largest difference in the DOM parameters obtained in case iii compared to case i was a 6% reduction in the magnitude of

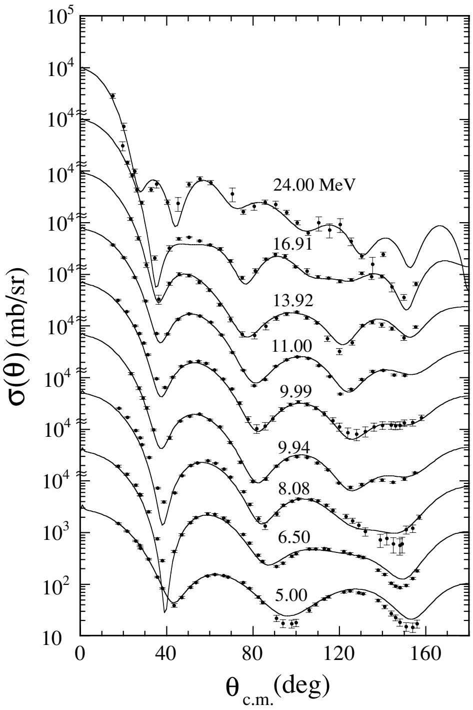

Figure 5. Plot of measured differential cross-section values for Sn(n, n) (solid circles). (See text for sources.) The solid curves are the calculations from the DOM. Compound-nucleus contributions were negligible for these energies. The data at 5.00, 6.50, 8.08 and 9.99 MeV are from the 'test database'.

the B_{HF} coefficient in the exponent in equation (3) for the Hartree–Fock term. Because of the form of the dispersion integral, the greatest differences between calculations with the DOMs of cases i and iii are at large negative energies, i.e., in the vicinity of the deepest hole states; this energy region ($E < -40$ MeV) was not investigated in the present work.

7. Predictions of bound-state properties

Because the volume and surface absorption terms (equations (4) and (5), respectively) are assumed to be symmetric for a large region around the Fermi energy, the DOM can easily be extrapolated to the negative energy regime. In this way, a DOM potential that is created by fitting to neutron elastic scattering data may be considered, in the negative energy regime, as a shell-model potential and used to describe neutron single-particle and single-hole bound states. In our study of the $n + {}^{120}\mathrm{Sn}$ system, the binding energies were calculated with our code BSEAUTO, which is based on a code provided to us by Johnson [25]. The occupation probabilities and spectroscopic factors were calculated using the code OCCUP reported in [26].

The DOM obtained for case i was used to calculate binding energies, spectroscopic factors and occupation probabilities for single-particle and single-hole states down to -15 MeV. In table 2, the results for particle states are compared with experimental values obtained from ${}^{120}\mathrm{Sn}(d, p){}^{121}\mathrm{Sn}$ and ${}^{120}\mathrm{Sn}(\alpha, {}^{3}\mathrm{He}){}^{121}\mathrm{Sn}$ stripping reactions. The results for hole states are compared to values obtained from ${}^{120}\mathrm{Sn}(d, {}^{3}\mathrm{H}){}^{119}\mathrm{Sn}$ and ${}^{120}\mathrm{Sn}({}^{3}\mathrm{He}, \alpha){}^{119}\mathrm{Sn}$ pickup

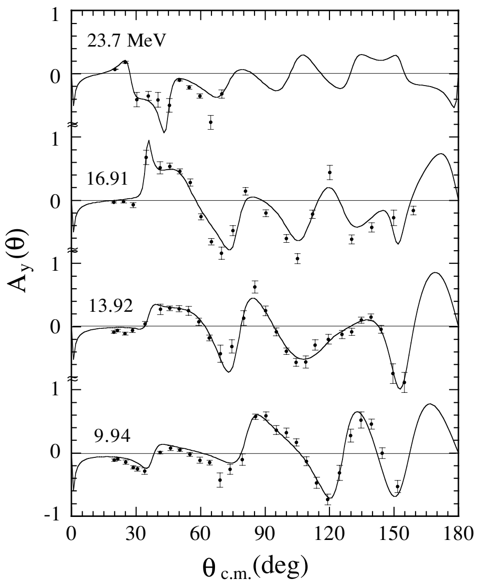

Figure 6. Plot of measured analysing-power values for Sn(n, n) (solid circles). (See text for sources.) The solid curves are the calculations from the DOM. The data are from the 'search database'.

Table 2. Properties of neutron single-particle and single-hole bound states.

State	Binding energy (MeV)		Spectroscopic factor		Occupation probability
	Prediction	Experiment	Prediction	Experiment	
Particle[a]					
$2f_{5/2}$	−0.576	−0.371	0.84	0.94	0.096
$3p_{3/2}$	−1.500	−2.490	0.82	0.56	0.079
$2f_{7/2}$	−2.661	−2.793	0.71	0.41	0.129
$1h_{11/2}$	−6.545	−6.165	0.61	0.38	0.255
Hole[b]					
$3s_{1/2}$	−8.404	−9.107	0.68	0.90	0.873
$2d_{3/2}$	−8.190	−9.131	0.66	0.45	0.965
$1g_{7/2}$	−9.902	−9.895	0.59	0.75	0.881
$2d_{5/2}$	−10.015	−10.276	0.61	0.68	0.881
$1g_{9/2}$	−14.304	−14.557	0.74	0.50	0.916

[a] Experimental information from [29, 30].
[b] Experimental information from [31–33].

reactions. The results using our case ii and case iii DOMs did not produce significantly different predictions and so they are not listed.

Moderate agreement is achieved with the measured binding energies and ordering of the bound states. Except for the two higher particle states, the binding energy predictions agree with experiment within 10%. We checked our determination of the Fermi energy by taking the average of the binding energies of the first particle and first hole states calculated with the present DOM. We obtained $E_F = -7.475$, which is close to the E_F of -7.639 MeV obtained with separation energies. The calculated spectroscopic factors agree with those observed in stripping and pickup reactions within about 35%. Most of the DOM values are larger than those extracted from stripping experiments. This result is not a surprise and is probably due to missing of some of the fragments in the analysis of the stripping spectral data.

8. Summary and conclusions

A dispersive optical model has been constructed for describing the $n + {}^{120}$Sn interaction from -15 to +80 MeV. The model was developed by searching for an optimum description to elastic scattering data for the range $9.9 < E < 24$ MeV and total cross-section data from 0.24 to 80 MeV, while applying constraints on the range of acceptance for the values of some of the parameters and using elementary energy dependences for the potential strengths. In addition, all the radius and diffuseness parameters were held constant with energy. Some parameters were adjusted to permit reasonable predictions for bound-state energies. Compound-nucleus calculations were performed and added to the DOM-calculated elastic scattering values for comparison to elastic scattering measurements below 4 MeV. Our final DOM was shown to provide exceptional agreement with measurements of $\sigma(\theta)$, $A_y(\theta)$ and σ_T data over the range $0.1 < E < 80$ MeV. This DOM was also used to predict bound-state quantities. Moderate agreement was found for binding energies and spectroscopic factors between our predictions and values derived from experiment.

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